

LAB VI: Evaluate Different Arithmetic Operations and Logical operations on two 32-bit data using ARM processor.

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Branch:		Section:	
S. No.	Name	Registration No.	Signature

Marks: ____/10

Remarks:

Teacher's Signature

Objectives:

1. Perform Addition and Subtraction of two 32-bit numbers using data processing addressing mode (with immediate data).
2. Perform Addition, Subtraction, and Multiplication of two 32-bit numbers using load/store addressing mode.
3. Perform the logical operations (AND, OR, XOR, and NOT) on two 32-bit numbers using load/store addressing mode.

Post Lab:

1. Give any examples of five arithmetic and logical instructions.
2. Differentiate between LDR and STR instruction.
3. Which of the following instructions is not valid
 - a) MOV R7,R2
 - b) LDR R1, =LABEL
4. Explain briefly condition codes (flags) of ARM processor.
5. Which condition codes (flags) is considered for the following branch instructions?
B Label
BEQ label
BLT label
6. If the registers r1,r2,r3 contain the values 10,20,30 respectively, what will be the value in register r4 after the execution of the following code segment?
ADD r4, r1, r3
SUBS r4, r2, r4
RSB r4, r1, r4